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Sono-optogenetics facilitated by a circulation-delivered rechargeable light source for minimally invasive optogenetics

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1. Introduction

What is the problem this paper is trying to solve?

In vivo optogenetic modulation requires invasive craniotomy and intracranial implantation of tethered optical fibers due to the limited tissue penetration of visible light.

Is this problem important?

Optogenetics offers neuron-type-specific neuromodulation with high precision. However, most of current interfaces not only perturbs endogenous neural activities but also leaves permanent damage of tissues.

What have others done to solve this?

1) Opsins with red-shifted activation spectra/2-photon stimulation

   → Partial removal of scalp and skull

2) Intracranial injection of upconversion nanoparticles

   → Intracranial delivery of photoluminescent agents into deep brain tissue

1. Introduction

What has this research done differently to solve this?

1) Replaced direct light illumination with focused ultrasound (FUS)
2) Intravenous light source delivery
3) Developed a light source based on mechanoluminescent nanoparticles

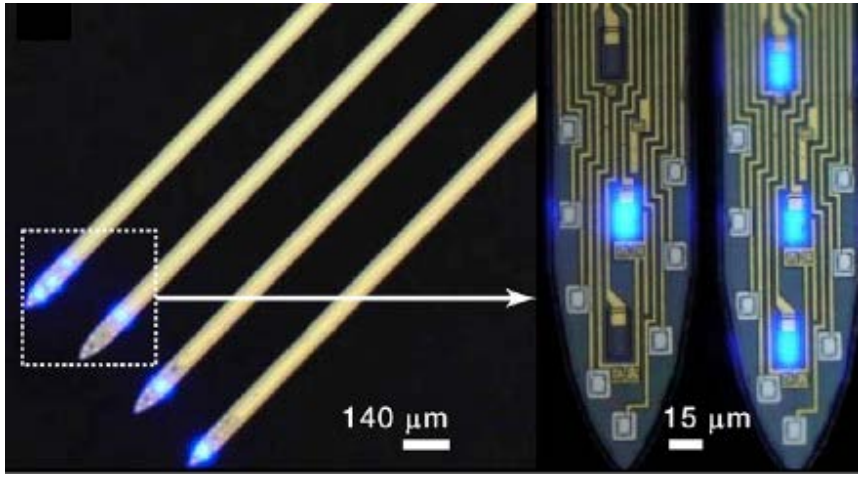
Results

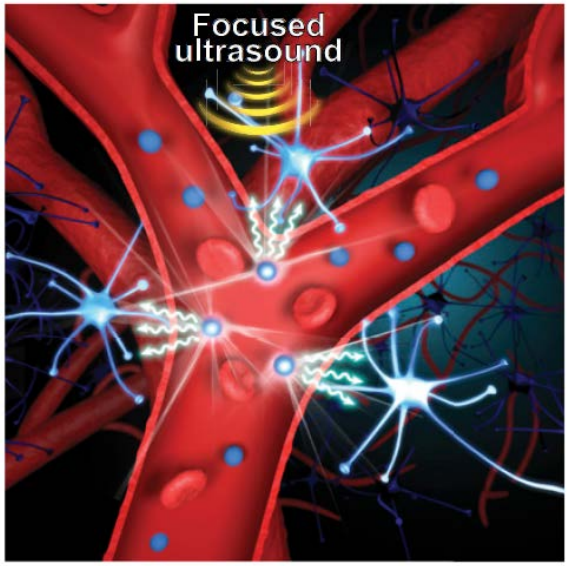
Sono-optogenetics: combination of FUS excitation and intravenous delivery of mechanoluminescent nanoparticles for fiber-free optogenetics through intact scalp and skull in live animals

1. Introduction

1. Optogenetic stimulation

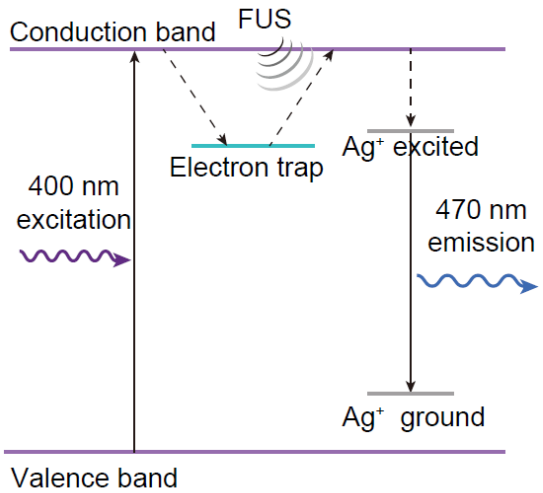
- **Mechanism**: use photosensitive ion channels or proteins expressed in genetically modified neurons
- **Advantages**: 1) high temporal precision 2) single-cell-type precision
- **Limitations:** 1) safety issues applying to humans 2) parasitic heat issues
- **Rising technologies and research directions**
  1. miniaturized μ-ILEDs
  2. mechanoluminescent particles

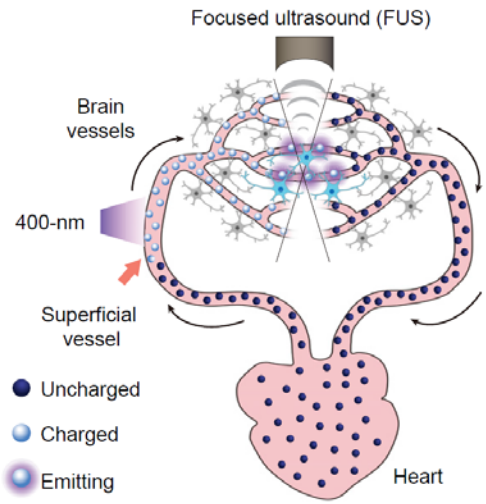




1. Introduction: Mechanism of sono-optogenetics

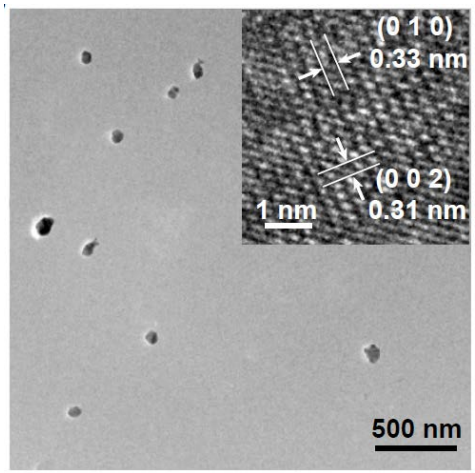
1. **Photoexcitation**: ZnS has a bandgap energy of 3.7 eV, leading to efficient absorption of ultraviolet light and excitation of electrons to the conduction band
2. **Defect-induced trapping**: Co2+ dopant ions in the ZnS matrix create defect states with a trap depth of 0.5 eV below the conduction band, which trap the excited electrons and store the photoexcitation energy without emission
3. **FUS-triggered detrapping**: when FUS is applied, the mechanical stress leads to charge separation in the piezoelectric ZnS matrix, effectively tilting the conduction band and making it easier for the trapped electrons to return to the conduction band
4. **Energy transfer**: Ag+ dopant ions receive the energy transferred from the detrapped electrons
5. **Photoemission**: produce a 470-nm emission light



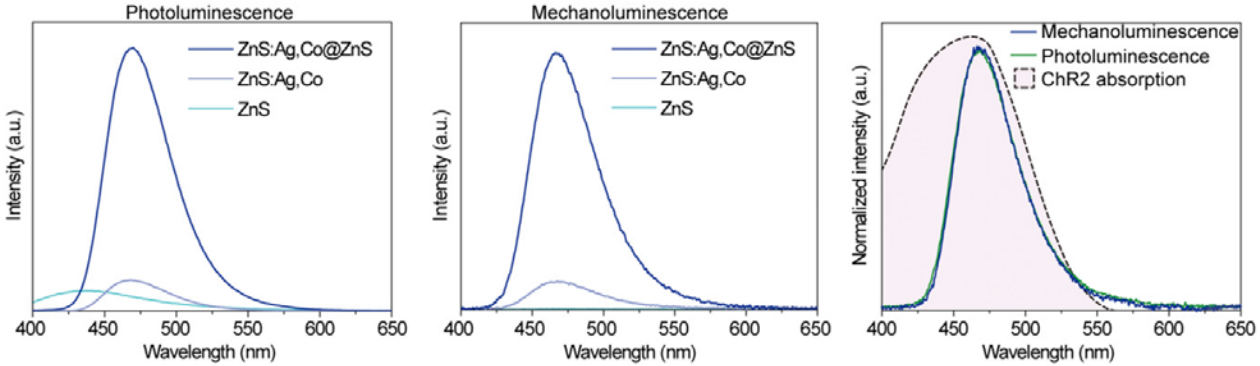


2. Characterization

1. **Morphological:** Core-shell ZnS:Ag, Co@ZnS nanoparticles have a spherical-like morphology with an average diameter of 86.6 (TEM image)



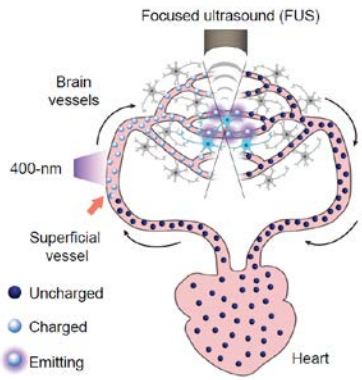
2. **Spectral**
   - 8.6-fold increase of photoluminescence
   - 9.2-fold increase on mechanoluminescence
     - compared to core-only ZnS:Ag nanoparticles
   - core-shell ZnS:Ag, Co@ZnS → 470 nm emission peak
     - Effective optogenetic activation of ChR2
     - core-only ZnS:Ag, Co → 430-nm emission peak
   - 400-nm photoexcitation and 470-nm emission characteristic
     - 200 um penetration depth (400 nm)



3. Experiment-1: Feasibility as use of rechargeable light sources

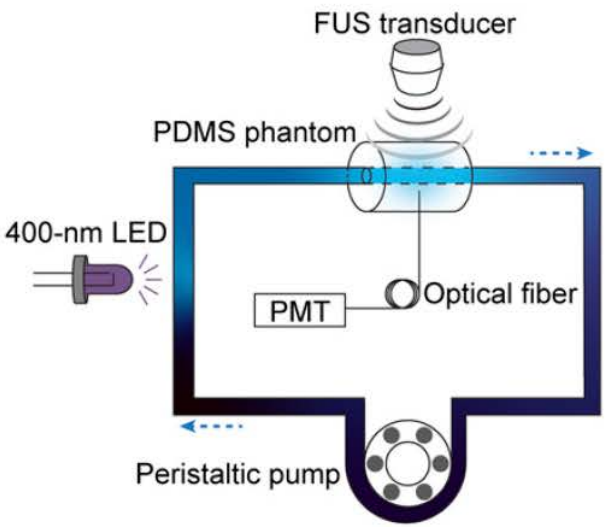
Artificial circulatory system

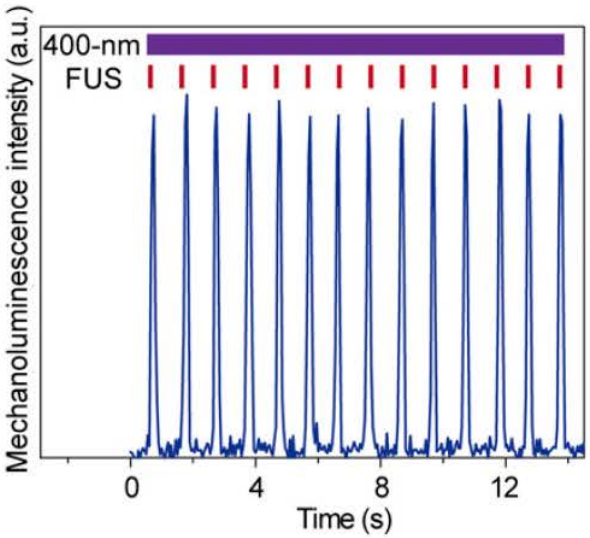
1. Cylindrical PDMS → mimic deep blood vessel
2. Tygon tubing at the both ends of PDMS tunnel → mimic blood stream
3. Peristaltic pump → mimic heart
4. 400-nm light source next to the Tygon tubing → recharge circulating nanoparticles
5. FUS transducer → excite the circulating nanoparticles inside the tunnel

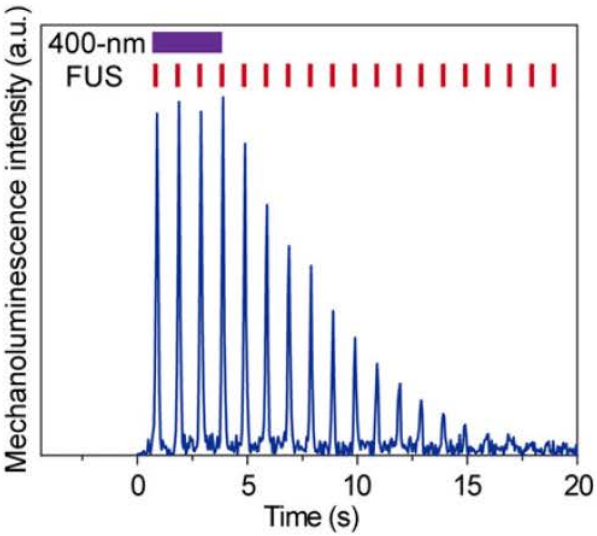


Measured the local light emission from circulating nanoparticles when FUS was applied at 1 Hz

1. Stable peak intensity of emitted light (5 % variation) → steady state of charging and discharging
2. When 400-nm photoexcitation light source was OFF → rapid decrease of peak intensity of the mechanoluminescence

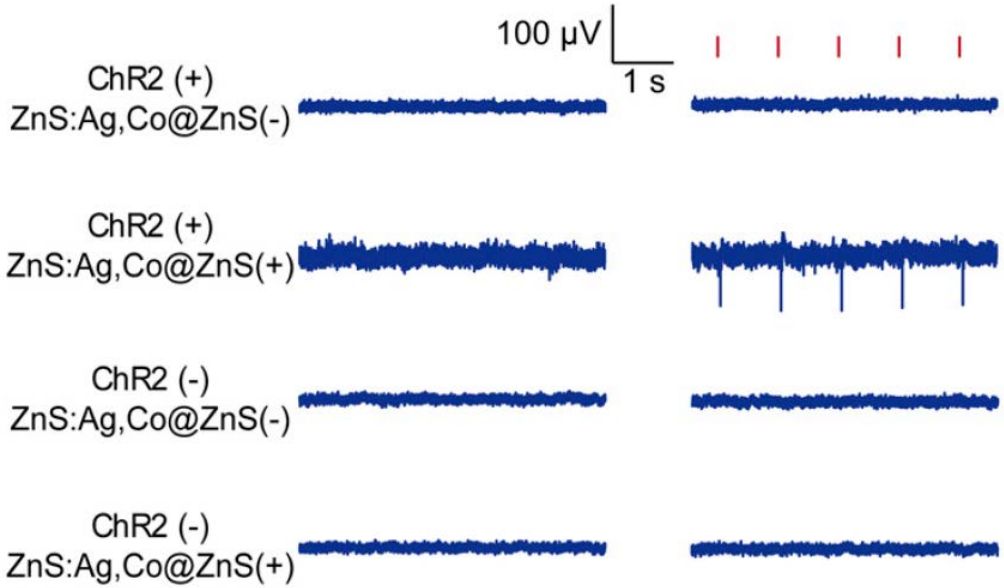


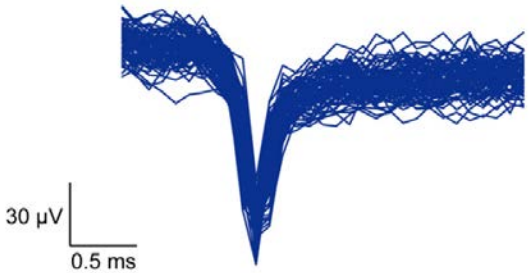


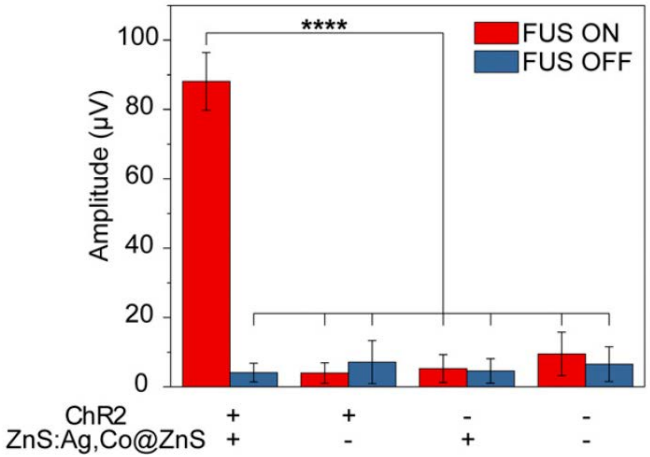


3. Experiment-2: Action potentials from spiking cells expressing ChR2 under FUS stimuli

- **Human embryonic kidney (HEK) cells transfected with ChR2**
- Extracellular recordings with MEA: periodic single-unit action potential only in the ChR2(+)/nanoparticle(+) group with FUS excitation
- Revealed typical waveforms of extracellular sing-unit spikes as negative peaks



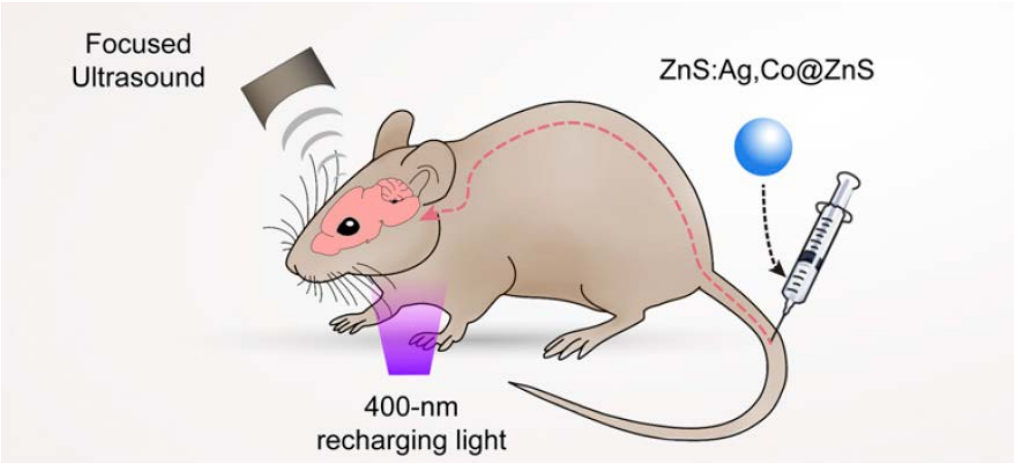


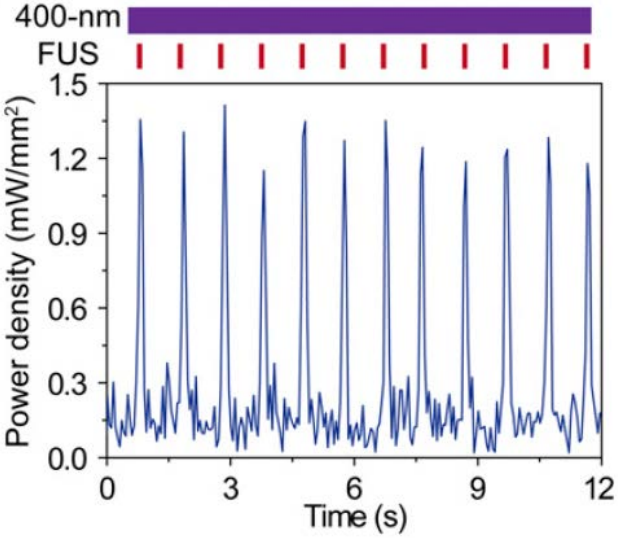


3. Experiment-3: Optogenetic stimulation of ChR2-expressing neurons in live animals

Whether ZnS:Ag, Co@ZnS nanoparticles in the intrinsic circulatory system in live animals allowed for optogenetic stimulation of ChR2-expressing neurons in the brain without craniotomy

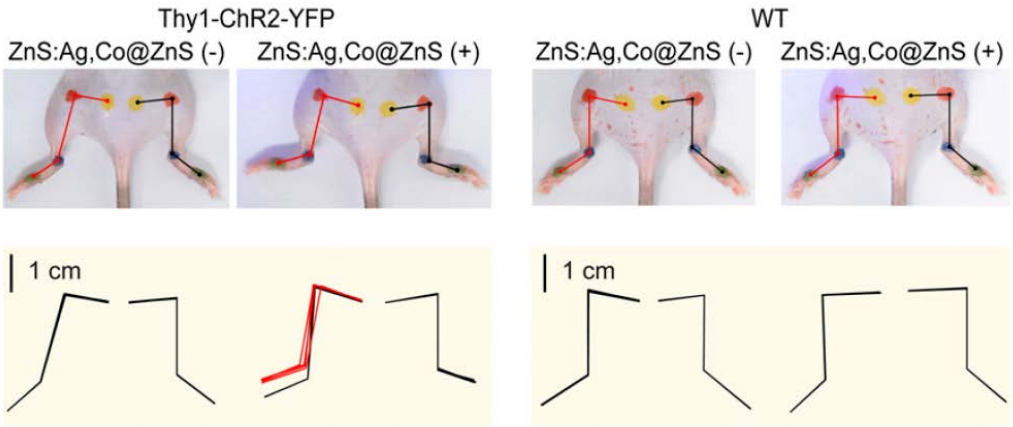
- Thy1-ChR2-YFP mouse under anesthesia
- In vivo luminescence measurements
  1. 8mg/mL in the blood circulation produced 470-nm light emission under FUS excitation with 1.2mW/mm^2
  2. Spiking probability stayed above 80 % of initial concentration for the first 0.5h after intravenous injection
     - ❖ Compared to 50 % before injection
  3. Temperature increase in the local brain was found to be <0.2 °C over 10s

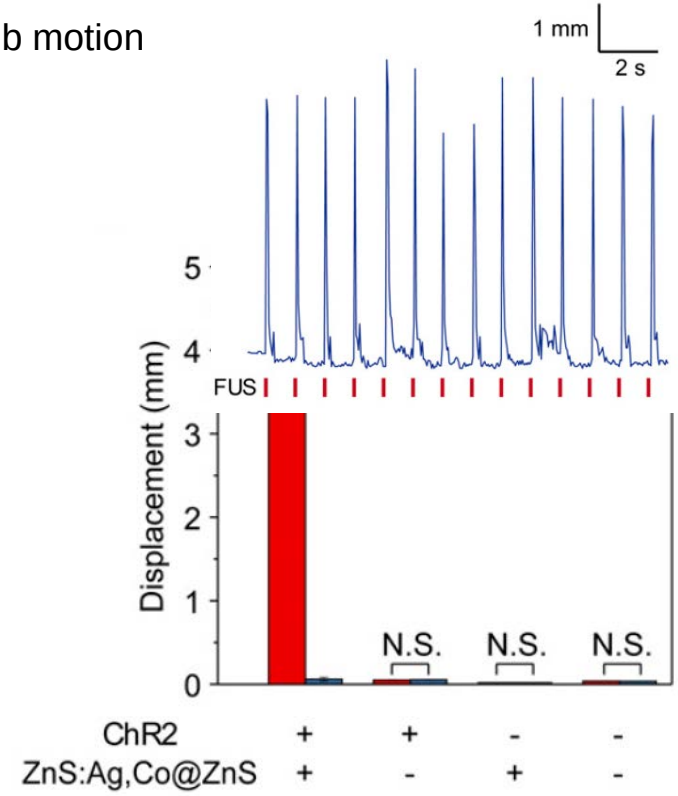




3. Experiment-4: Activation of unilateral limb movement

- **Activation of unilateral limb movement by focusing ultrasound to the secondary motor cortex (M2) subserved by blood circulation carrying chaged nanoparticles in the right hemisphere though intact scalp and skull**
- Thy1-ChR2-YPF mouse with CHR2 expressed in the M2 demonstrated hindlimb motion synchronized with FUS excitation
- Wild-type mouse w/o ChR2 expression demonstrated no hindlimb motion
  - → FUS stimuli alone were unable to evoke limb movements
- Unilateral motion only in the left hindlimb of the ChR2 mouse
  - → Contralateral to the focus of ultrasound in the right hemisphere
- Reproduced in a group of 3 Thy1-ChR2-YFP mice





4. Conclusions

Summary

1) Minimally invasive in vivo optogenetic stimulation in live mouse brain using a sono-optogenetic stimulation in live mouse brain
2) Advantage of the intrinsic circulatory system to deliver nanosized light sources
3) Efficacious ChR2 activation in vitro and neuromodulation with motor behavior changes in vivo

Challenges

1) Toxicity concerns
2) Body-mounted systems for generation of FUS
3) Potential for parasitic heating